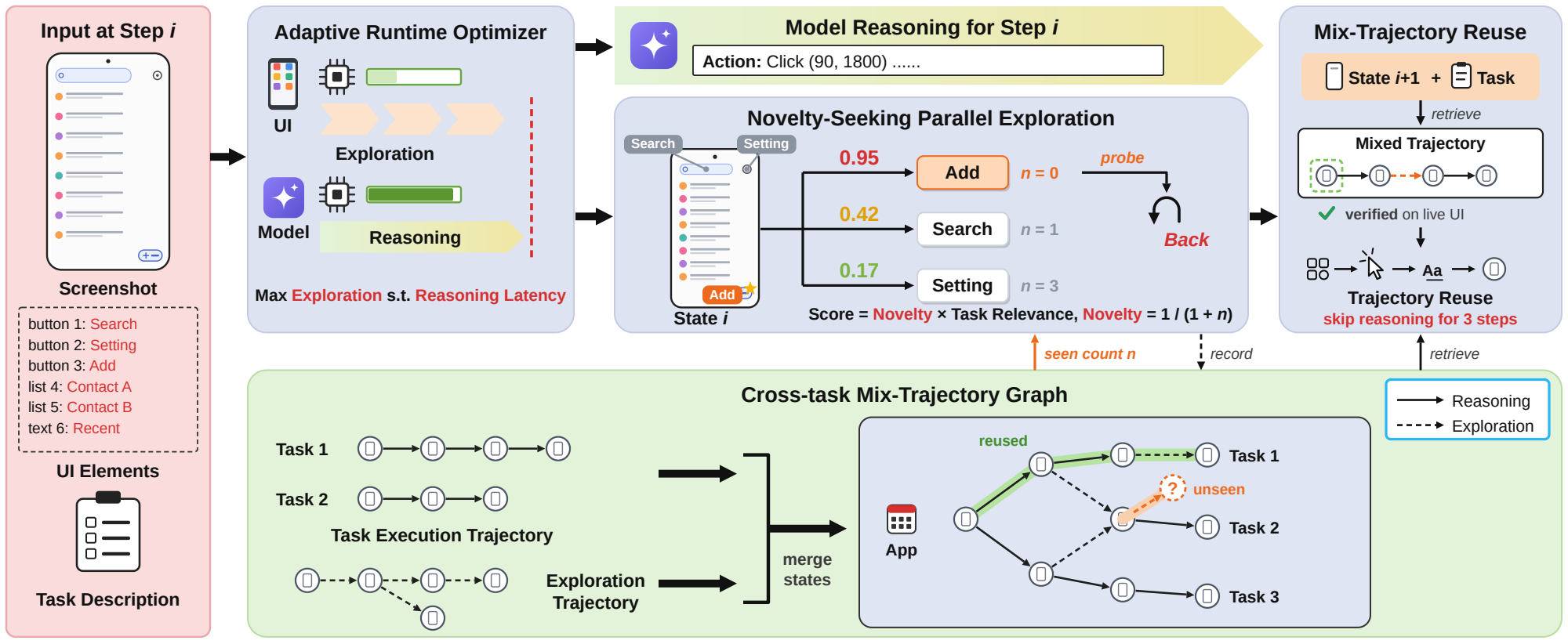

Input at Step i
Screenshot
button 1: Search
button 2: Setting
button 3: Add
list 4: Contact A
list 5: Contact B
text 6: Recent
UI Elements
Task Description
Adaptive Runtime Optimizer
UI
Exploration
Model
Reasoning
Max Exploration s.t. Reasoning Latency
Model Reasoning for Step i
Action: Click (90, 1800) .....
Novelty-Seeking Parallel Exploration
Search
Setting
Add
State i
0.95
Add
n = 0
probe
0.42
Search
n = 1
Back
0.17
Setting
n = 3
Score = Novelty × Task Relevance, Novelty = 1 / (1 + n)
Mix-Trajectory Reuse
State i+1 + Task
retrieve
Mixed Trajectory
verified on live UI
Aa
Trajectory Reuse
skip reasoning for 3 steps
seen count n
record
retrieve
Cross-task Mix-Trajectory Graph
Task 1
Task 2
Task Execution Trajectory
Exploration Trajectory
merge states
App
reused
unseen
Task 1
Task 2
Task 3
Reasoning
Exploration